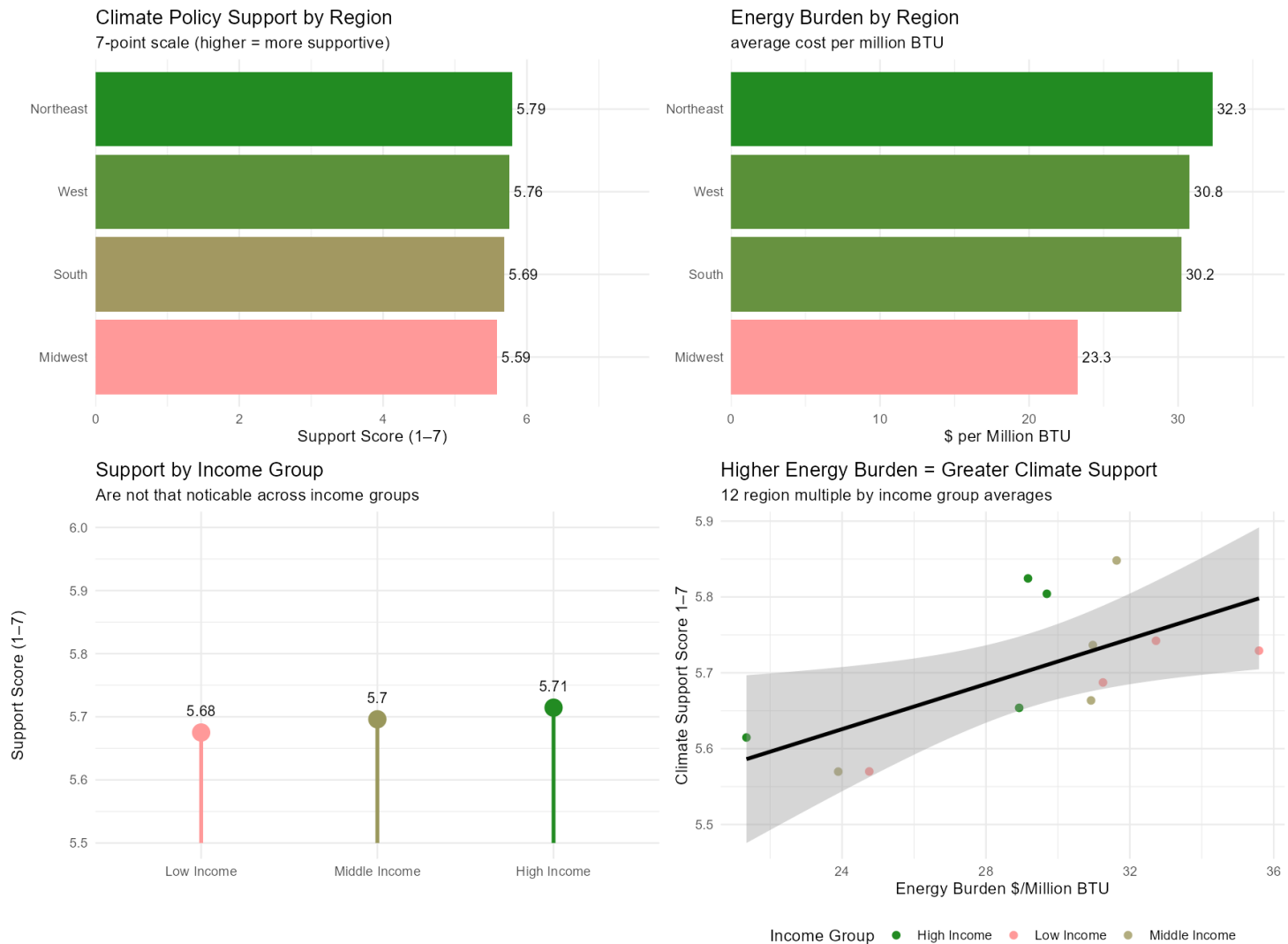




Energy Costs and the Political Climate of Climate Support

The Northeast's high climate policy support correlates with its high energy burden. However, The rest of the U.S. it's the opposite.

Climate policy support varies across U.S. regions. When you first look at it, it makes sense. The Northeast is more liberal and tends to be wealthier on average compared to conservative areas. Liberals are also more likely to hold a baccalaureate degree. However, there are many more unobserved variables that affect these regions of the U.S. and their support for climate policy. The Northeast leads the country in climate policy support (5.79 out of 7 using ANES data metrics) and also pays

the most for energy (32.30 USD per million BTU). The Midwest, on average, supports climate policies less, at around 5.58. The Midwest only pays 23.30 USD per million BTU. At first glance, this makes economic sense: regions that feel energy costs the most usually demand cleaner, more stable alternatives (e.g., nuclear energy, which produces the most amount of energy for the least amount of “raw ingredients,” excluding the startup costs). The regression shows a real relationship, a one-unit increase in energy burden predicts a statistically significant increase in climate support (coefficient = 1.045e-03, $p < 2e-16$, which is highly significant). But the R-squared is 0.035, which means that this is not the best fit (other factors are explaining most of the variation). Energy burden and income together explain only 3.5% of the variation in climate attitudes. There might be more unexplained variables that can explain the re-

lationship better.

Education is one of the biggest factors that determine if an individual supports climate policies. The Northeast is not just the region with the highest energy costs, but it is also the most education-dense region in the country, home to the highest concentration of universities, research institutions, and college-educated people. Boston, New York, Philadelphia, and New Haven show the highest support for climate policies. They have a highly educated population with a deep understanding of green energy and climate policy. Because these residents face high heating costs, they have an advantage in recognizing that climate initiatives can be subsidized by the government as well, which reduces costs to the consumer. When the education variable is left out of the model, the energy burden coefficient absorbs some of the education effect, which is known as omitted variable

bias. Education is positively correlated with both energy burden (denser, more urbanized regions tend to have both) and climate

support, so omitting it overstates the effect of energy burden on whether people support climate policies. The true effect of en-

ergy costs on climate support is probably smaller than the coefficient shows.

Call:

```
lm(formula = avg_climate_support ~ energy_burden + income_group,
    data = indiv_data)
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Residuals:

Min	1Q	Median	3Q	Max
-0.22238	-0.07442	-0.05171	0.09194	0.16498

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	5.690e+00	1.483e-03	3835.899	< 2e-16	***
energy_burden	1.045e-03	4.774e-05	21.889	< 2e-16	***
income_groupLow Income	-3.976e-02	2.802e-03	-14.191	< 2e-16	***
income_groupMiddle Income	-1.561e-02	2.462e-03	-6.339	2.36e-10	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08691 on 18492 degrees of freedom

Multiple R-squared: 0.03484, Adjusted R-squared: 0.03468

F-statistic: 222.5 on 3 and 18492 DF, p-value: < 2.2e-16

Across all three income groups, climate support barely moves: the low income household average is 5.68, middle income is 5.70, and high income is 5.72. The regression uses high income households as the reference category. Low income respondents score 0.039 points below them ($p < 2e-16$) and middle income respondents score 0.015 points below ($p < 2e-16$). These are not very meaningful distinctions on a 7-point scale. The assumption is that climate policy support rises steeply with wealth. While income and education are often correlated, income itself is not a good predictor of climate support. Maybe the reason why college-educated people support climate policies is because of the social “scene” that education provides (e.g., environmentally related protests are normal on campuses).

The regional pattern in Figure 2 shows that there are unobserved variables that are affecting the energy burden. The South has the second highest energy burden (30.20 USD/million BTU) — higher than the West

and bigger than the Midwest. Yet, it ranks lower in climate support at 5.69. If energy costs directly produced support for climate policies, then the South should support climate policies like the Northeast. The gap between the South’s energy burden and its climate support, compared to the Northeast, is just the kind of residual that an omitted variable like the density of a population that is highly educated would explain. Northeastern households pay more for energy and live in environments where climate concerns are normal. Southern households pay almost as much, but they are less likely to be college-educated. Hence, less likely to support climate policies.

The regression was run on individual households in the U.S., about 18,495 observations, with climate support assigned as the region-by-income group average from the ANES dataset. Since the two datasets do not share individual identifiers, the results are based on only 12 group averages. The findings look more consistent than they re-

ally are. Which hides the true variation between individuals. The results clearly show the trend of the relationship, but the size of the effect, I think, is less precise. A better-specified model should include college attainment rates as a control, which isolates the effect of energy costs by adjusting for the differences in education across the region. Assuming that the current model is positively biased, the coefficient would likely reduce — which would make it easier to figure out what is going on in the Northeast. Adding more variables to the regression might also increase multicollinearity.

High energy bills do not automatically make people support climate policy. While the Northeast has both high energy costs and high climate policy support, the South has high energy costs without high support for climate policies. I believe the difference is the density of college-educated people, the large number of colleges in that area, and other correlated variables. 🐼